SPEAR3 Legacy LLRF

Tuner Control System Technical Analysis

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This document reverse-engineers the mechanical cavity tuner loop *as deployed on SPEAR3*, from the legacy `spear-rf-code-legacy/rfApp/` source tree. Two things are easy to get wrong when reading it. First, the architecture is a dual loop, but only the inner phase-regulation loop is active on SPEAR3: the outer voltage loop is PEP-II inheritance and its phase offset is fixed at zero. Second, the **AB 1746-HSTP1 is still the in-service controller**; the Galil DMC-4143 has been built and tested and goes online, driven by the LLRF9, only when the LLRF upgrade completes. The Open Items register lists the points that still require field verification.

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Open Items

Each entry below is a statement in this document that rests on an operator screen, a vendor note or an undocumented convention rather than on a file in the repository. The page link returns to the statement.

ID	Item	Required evidence	Needed before	Page
O1	The **Tuner Inputs & Constants** operator panel is the only stated source for the nominal values of every parameter that is commented out in the database source.	Archive the screen file (or an equivalent configuration export) in the repository, and record which values autosave actually restores on the running IOC.	Any claim that the tuner is functional as configured	p. 30
O2	The mm-per-degree conversion is quoted as a **range**, 0.020–0.030 mm/°, not a single commissioned number.	The per-cavity commissioned value, or the measurement that established the range.	Quantitative loop-gain analysis	p. 6
O3	The **IQA phase calibration** that defines the zero of `LOAD:ANGLE:ERR` has no recorded procedure or last-performed date.	A written calibration procedure and the date it was last run on each cavity.	Trusting that `ERR` = 0 means resonance	p. 8
O4	Whether **autosave** is actually enabled for the tuner parameter set on the production IOC.	The autosave request file list and a restart test.	Any statement about behavior after an IOC restart	p. 15
O5	The **loop gain** `CAVTUNR:LOOP:GAIN` is stated to run at unity, but the value is operator-settable and not archived.	The as-running value, read from the live IOC.	Reproducing the loop bandwidth figures	p. 12
O6	The **Galil DMC-4143** replacement has test moves recorded under an EPICS control by M. Dunning; in service LLRF9 Unit 1 will drive it directly, and that configuration has no commissioning record or cutover plan yet.	A commissioning test with the LLRF9 driving the Galil, including turn-on from a detuned tuner position, and the cutover criteria.	Scheduling the controller swap	p. 21
O7	The inner-loop gain Eq. (8.3) reuses the *park*-mode degrees-per-kHz constant `0.00004726891`, which differs from the physical load-angle conversion by about a factor of five.	Confirmation of which conversion the inner loop is correctly analyzed with, ideally from a measured step response.	Any quantitative stability margin	p. 20

Part I
What and Why

1 Executive Summary

The SPEAR3 tuner control system keeps each cavity at its resonant frequency by mechanically adjusting a tuner plunger in response to measured phase error. The primary disturbance it compensates is **beam-loading-induced detuning** — the reactive component of beam current shifts the cavity's apparent resonant frequency in proportion to beam current. A secondary, slower disturbance is thermal drift from RF heating.

The system was designed as **two nested feedback loops**:

Loop	What it regulates	Controller type	Rate	Status in SPEAR3
Inner	Phase error (probe − forward)	Proportional → stepper motor	0.5 Hz	**Active**
Outer	Cavity voltage (strength)	Leaky integrator → phase offset	0.5 Hz	**Disabled by default**

In SPEAR3 as deployed, only the inner loop is active. It drives the cavity to the resonant frequency calibrated during a no-beam reference procedure. The outer voltage loop exists in the code but its output is structurally clamped to zero (see §5.3). The PARK mode frequency-domain path is PEP-II functionality, permanently disabled.

Key numbers:

- 0.5 Hz loop rate (one cycle every 2 s), 4 cavities, station prefix `SRF1:`
- Stepper motor: 3.81 mm/s, 0.003175 mm/step, range −29.5 to +18 mm
- Phase deadband: 0.25° (design), conversion: 0.020–0.030 mm/° (nominal operational range per [R15]) [O2], max delta: 1 mm/cycle

---

2 Physics: Cavity Detuning and Beam Loading

Sources: [R18].

2.1 Why the Tuner Exists

An RF cavity has a resonant frequency f_0 determined by its geometry. When driven off-resonance, the generator must supply extra reactive power. The mechanical tuner adjusts cavity geometry to keep f_0 aligned with the drive frequency (476 MHz at SPEAR3), minimizing reflected power and maximizing efficiency.

Two phenomena shift the effective resonant frequency during operation:

Beam loading (primary, fast): When stored beam traverses the cavity, the reactive component of the beam-induced current shifts the apparent resonant frequency by:

$$\Delta f_{beam} = -\frac{f_0}{2} \cdot \frac{R}{Q} \cdot \frac{I_{b1} \sin \phi_s}{V_{gap}} \tag{2.1}$$

where $I_{b1} = 2I_b$ is the *fundamental* component of the bunched beam current (I_b being the DC average), ϕ_s is synchronous phase, $R/Q \approx 117\,\Omega$ is the cavity geometric shunt-impedance ratio, and V_{gap} is gap

voltage. Because R/Q depends only on cavity geometry, Eq. (2.1) is independent of Q_L and of the coupling factor; the equivalent loaded form is $\Delta f_{beam} = -(f_0/2Q_L)\, I_{b1} R_L \sin\phi_s / V_{gap}$ with $R_L = R_s/(1+\beta)$ and $Q_L \approx 7000$. This detuning is proportional to beam current and changes dynamically as beam is injected, decays, or is lost. At SPEAR3 operating conditions ($I_b \sim 100$–500 mA) beam loading detunes the cavity by tens of kHz — about -36 kHz at 500 mA.

Thermal drift (secondary, slow): RF heating causes slow mechanical deformation of the cavity body, shifting the resonant frequency on timescales of minutes to hours. At SPEAR3's operating gradient this is much smaller than beam loading.

2.2 The Load Angle

For a driven cavity with beam loading, define:

- ϕ_L = load angle = phase of cavity field relative to generator
- ψ = forward coupler phase relative to cavity probe

The optimal detuning $\Delta\omega_{opt}$ for minimum generator power is the same quantity expressed in angular frequency:

$$\Delta\omega_{opt} = -\frac{\omega_0}{2Q_L} \cdot \frac{I_{b1} R_L \sin\phi_s}{V_{gap}} \tag{2.2}$$

This corresponds to a target load angle ψ_{target} where the generator operates most efficiently; for SPEAR3 at 500 mA that target is $\psi_L \approx -46^\circ$ (Doc P).

2.3 How the Error Signal Maps to Detuning

The inner loop measures:

$$\text{error} = \phi_{probe} - \phi_{fwd} + \theta_{offset} \tag{2.3}$$

In Eq. (2.3), with $\theta_{offset} = 0$ (as in SPEAR3), `error = 0` means $\phi_{probe} = \phi_{fwd}$, which corresponds to **resonance** (zero detuning). The tuner drives toward this condition.

With a nonzero θ_{offset}, the tuner drives to $\phi_{probe} - \phi_{fwd} = -\theta_{offset}$, applying a deliberate detuning. The exact relation is $\tan\psi_L = 2Q_L\, \Delta f / f_0$, so for the small angles the loop actually works over:

$$\Delta f \approx \frac{f_0}{2Q_L} \cdot \frac{\pi\,\theta}{180^\circ} \qquad \text{i.e.} \qquad \Delta f \approx \frac{f_0\,\theta}{114.6\,Q_L} \tag{2.4}$$

At $Q_L \approx 7000$ this is about 0.6 kHz per degree of load angle.

2.4 The IQA Calibration Requirement — Why the Target Is Zero

The inner loop drives `LOAD:ANGLE:ERR` $\rightarrow$ `0`. But the raw IQA measurements include arbitrary phase offsets from cable lengths, directional coupler geometry, and signal conditioning hardware. The measured phase difference at true resonance is generally *not* zero.

A calibration step establishes the zero reference:

1. With **no beam** (so beam loading does not shift the resonant frequency), operators sweep the tuner through resonance
2. At resonance (confirmed by minimum reflected power or maximum stored energy), the EPICS IQA phase offset parameters are set so that `PROBE:PHASE.L` $-$ `FRWD:PHASE.L = 0`
3. From that point forward, `LOAD:ANGLE:ERR = 0` means resonance; any nonzero value ϵ means detuning by $\Delta f \approx f_0\,\epsilon/(114.6\,Q_L)$ (Eq. (2.4))

EPICS panel parameters for this calibration: The per-signal offsets described in step 2 are set via the **"Probe Phase Offset"** and **"Frwd Phase Offset"** parameters on the Tuner Inputs & Constants panel [R15]. These are the operator-accessible knobs that translate the raw IQA `.L` measurement to zero at resonance. Both offsets are determined during tuner configuration.

Why `.L` rather than `.VAL`? Both `PROBE:PHASE` and `FRWD:PHASE` have a `.VAL` field with an individual per-signal phase offset subtracted (for beam physics applications). The `.L` field is the raw input before this offset. Using `.L` means the load angle calibration is set once at the system level (both `.L` traces zeroed at resonance) rather than needing to track per-signal offsets.

Practical implication: If the IQA phase calibration is incorrect or has drifted (hardware swap, cable re-routing), the tuner drives the cavity to the wrong frequency [O3] — it minimizes the measured phase difference, but that measured zero no longer corresponds to the true resonance. The system has no independent check of resonance other than this phase measurement.

---

3 Dual-Loop Architecture Overview

Sources: [R1], [R5], [R8], [R9], [R10], [R13].

3.1 The Two Loops

The system was designed with two nested feedback loops. The **inner loop** (phase regulation) runs once every 2 s (0.5 Hz) and drives a stepper motor to null the measured phase error. The **outer loop** (voltage regulation) is a slow integrator that adjusts the inner loop's phase setpoint to maintain cavity voltage. Figure 1 shows how the two nest.

In SPEAR3 as deployed: $\theta_{offset} = 0$ always. The outer loop exists in the code architecture but defaults to disabled on a fresh start without autosave (see §5.3). In normal operation with autosave restore the outer

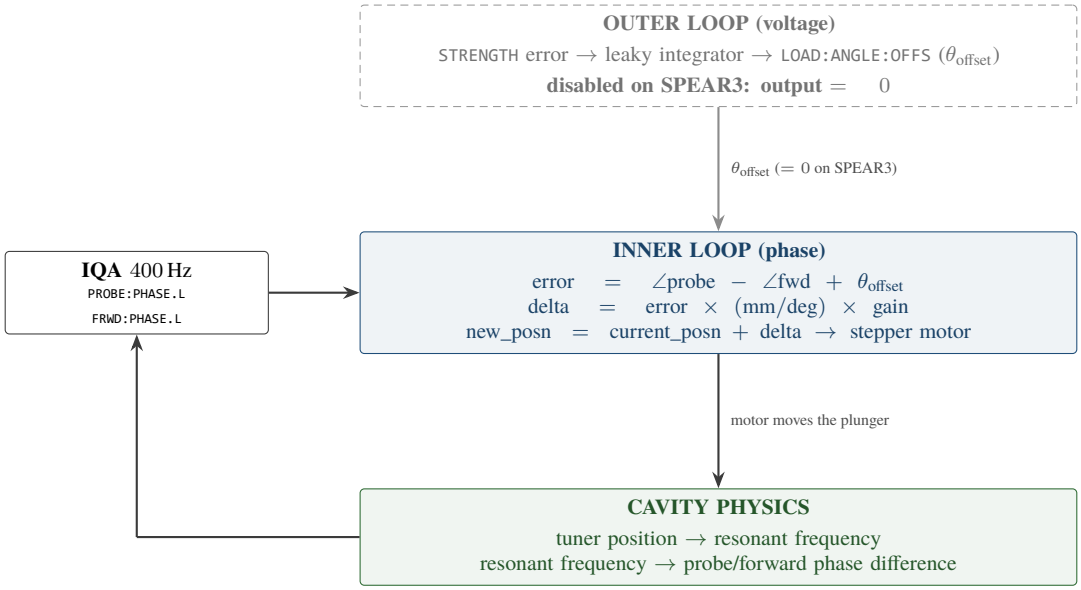


Figure 1: The dual-loop tuner architecture: an inner phase-regulation loop closed on the load angle error, and an outer voltage-regulation loop that biases its setpoint. On SPEAR3 only the inner loop is active.

loop can be functional. The tuner operates in pure resonance-seeking mode — zero phase error means resonance.

3.2 Station Configuration

Item	Value	Source
Station prefix	`SRF1:`	`rf_stn_4CV.substitutions,v`
Cavity prefix	`{STN}:CAV{N}` (N=1..4)	`rf_cav.db,v`
Number of cavities	4 (independent tuner per cavity)	`rf_tuner_loop.st,v`
RF frequency	476 MHz	`rf_cav.db,v`
Loop rate	0.5 Hz — one cycle every 2 s (SCAN="2 second") [R9]	`rf_stn_cav.db,v`

3.3 Station State Machine Context

The tuner loop runs in every station state except OFF [R1]. It moves the tuners in TUNE and `ON_CW` while the klystron forward power is above its minimum, holds in `ON_FM`, and in OFF moves them only to a home position on request. The state codes (from `rf_station_state.h,v`) are:

State	Code	Tuner Behavior
OFF	0	Loop off; tuners move only to the ON or PARK home position when homing is requested
PARK	1	PEP-II frequency-error path, without the forward-power check — disabled in SPEAR3
TUNE	2	Active control, as in `ON_CW`, while the klystron forward power is above its minimum
ON_FM	3	Loop holds (`ON_FM` status); PEP-II state
ON_CW	**4**	**Active control — phase-domain path**

3.4 Software Components

Figure 2 shows how the three software layers — the EPICS database, the SNL state machine and the motor record — hand the measurement down to the hardware.

3.4.1 Galil controller status

Per the system owner: the Galil DMC-4143 control **has not been installed and is not in operation**. It has been built and tested on the existing cavity motors under an EPICS control by M. Dunning, and the moves worked fine. In service, LLRF9 Unit 1 will run the four tuner loops and drive the Galil directly through its built-in Galil driver, so that EPICS control is not carried forward. The controller is expected to go online **when the current LLRF upgrade project is finished**. It has *not* replaced the Allen-Bradley hardware.
The AB 1746-HSTP1 stepper controller remains the in-service hardware. Everything in this document describing present-day tuner behavior refers to the AB path; Galil material is forward-looking.

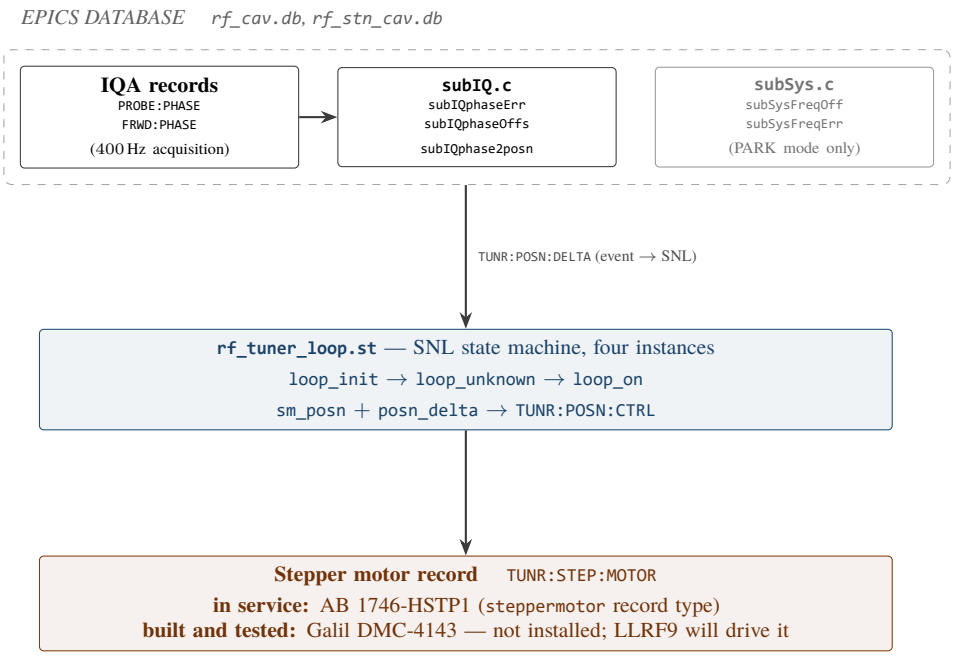


Figure 2: Tuner signal chain from the cavity probe and forward coupler through the IQA modules to the stepper motor record.

Part II
How It Works

4 Inner Loop: Phase Regulation

Sources: [R1], [R3], [R8].

This section traces the signal from IQA measurement through to motor command — the active control path in SPEAR3.

4.1 Step 1 — Phase Error Measurement

C function: `subIQphaseErr` — `rfApp/src/db/subIQ.c,v` line 566 [R3]
EPICS record: `$(S):$(C)LOAD:ANGLE:ERR`

```
psub->val = psub->l = psub->a - psub->b + psub->c + psub->d;
if      (psub->val < -180.0) psub->val += 360.0;
else if (psub->val >  180.0) psub->val -= 360.0;
```

$$\boxed{\text{LOAD:ANGLE:ERR} = \phi_{probe} - \phi_{fwd} + \theta_{offset} + 0} \tag{4.1}$$

Input	PV	Description
INPA (ϕ_{probe})	`PROBE:PHASE.L`	Unadjusted probe phase (degrees)
INPB (ϕ_{fwd})	`FRWD:PHASE.L`	Unadjusted forward phase (degrees)
INPC (θ_{offset})	`LOAD:ANGLE:OFFS`	Target phase offset (= 0 in SPEAR3)
INPD	—	Spare, always 0

Result wrapped to $[-180°, +180°]$. Alarm limits: `HIHI/LOLO` = $\pm 5°$ (MAJOR).

Load angle error limit (Ld Angle Err Limit): The operator panel exposes a separate parameter **"Ld Angle Err Limit"** (nominal **0.5**° per [R15]) that is distinct from the $\pm 5°$ EPICS HIHI/LOLO alarms. When `|LOAD:ANGLE:ERR|` exceeds this limit, the outer loading angle offset integrator (§5.2) is **frozen** — no further offset accumulation occurs until the error returns within bounds. This provides a guard against the outer loop winding up during large transient detuning events.

Note on `.L` field: Uses unadjusted (pre-offset) phase from IQA records. The `.VAL` field has a per-signal phase offset subtracted for beam physics applications — that subtraction is separate from the load angle target. See §2.4 for the calibration requirement that makes `error = 0` correspond to resonance.

```
grecord(sub,"$(S):$(C)LOAD:ANGLE:ERR") {
    field(INAM,"subIQinit")
    field(SNAM,"subIQphaseErr")
    field(INPA,"$(S):$(C)PROBE:PHASE.L  NPP MS")
    field(INPB,"$(S):$(C)FRWD:PHASE.L   NPP MS")
    field(INPC,"$(S):$(C)LOAD:ANGLE:OFFS.VAL NPP NMS")
    field(EGU,"deg")
    field(HOPR,"10")    field(LOPR,"-10")
    field(HIHI,"5")     field(LOLO,"-5")
    field(HHSV,"MAJOR") field(LLSV,"MAJOR")
}
```

4.2 Step 2 — Phase Error to Position Delta

C function: `subIQphase2posn` — `rfApp/src/db/subIQ.c,v` line 586 [R3]
EPICS record: `$(S):$(C)TUNR:POSN:DELTA`

```
// Select error source based on station state
if (state == STATION_PARK) error = G;  // park freq error (PEP-II)
else                       error = F;  // load angle error (SPEAR3 ON_CW)

gain = clamp(gain, 0, 1);  // CAVTUNR:LOOP:GAIN

if (|error| > deadband_B):
    delta = clamp(error * C_mm_per_deg * gain, -D_mm, +D_mm)
else:
    delta = 0
```

$$\boxed{\Delta x = \text{clamp}\big(\epsilon \cdot C_{mm/deg} \cdot G_{loop},\ -D,\ +D\big) \quad \text{if } |\epsilon| > B} \tag{4.2}$$

Input	PV / Value	Design Value	DB Status
A	`STN:STATE:RBCK` (station state)	—	wired
B	deadband (degrees)	0.25°	commented out
C	mm per degree	0.020–0.030 mm/° (nominal range per [R15])	commented out
D	max delta per cycle	1 mm	commented out
E	`CAVTUNR:LOOP:GAIN`	0–1; nominal **1.00** (always run at unity per [R15]) **[O5]**	wired
F	`LOAD:ANGLE:ERR`	degrees	wired
G	`FREQ:ERR` (PARK only)	degrees	wired (inactive)

! Configuration Gap: INPB, INPC, INPD are all commented out in `rf_cav.db,v`. At startup these default to 0, making C=0 (delta always 0) and D=0 (output clamped to 0). The tuner is non-functional without autosave restore or operator initialization.

4.3 Step 3 — SNL Motor Command Integration

Source: `rfApp/src/rf_tuner_loop.st,v` — the `loop_on` state, 0.5 Hz LOOP:READY event handler

After the IQA chain computes `POSN:DELTA`, the SNL sequencer integrates it into an absolute motor position:

```
pvGet(sm_posn)                          // current motor position (from .RBV)
pvGet(posn_delta)                       // phase→position result

// MOTOR SANITY CHECK: Did the motor reach last commanded position?
if (|posn_ctrl − sm_posn| > sm_rdbd) AND (prev_loop_ctrl == ON):
```

```
    loop_status = LOOP_SM_CTRL_STATUS       // missed previous target
    prev_loop_ctrl = OFF                    // suppress until re-enabled

// Compute new absolute target
posn_new = sm_posn + posn_delta             // integrate onto current position
posn_new = clamp(posn_new, sm_drvl, sm_drvh) // enforce travel limits

// Write only if position changed or loop just re-enabled
if (posn_ctrl != posn_new) OR (prev_loop_ctrl != ON):
    posn_ctrl = posn_new
    pvPut(posn_ctrl)                        // → FLNK → motor record
prev_loop_ctrl = loop_ctrl
```

Key design choice: The SNL uses `sm_posn` (motor `.RBV` — actual position) as the base for each cycle, not the last commanded `posn_ctrl`. This means the inner loop is truly proportional — each cycle's motor command is: $x_{new} = x_{actual} + \Delta x$. The position integration is *implicit* through new commands building on the actual position.

Outer loop trigger (coupling to §5): After the motor update, the SNL also fires the outer voltage integrator:

```
if (station_state != STATION_PARK) AND (dmov_meas_count >= 1):
    pvPut(phase_offset_proc)     // writes PROC to LOAD:ANGLE:UNADOFFS
```

This `phase_offset_proc` write is the only mechanism that triggers the outer loop. It only fires when the motor was idle (`DMOV=1`) and the inner loop completed a full update. See §5 for the outer loop details.

---

5 Outer Loop: Voltage Regulation

Sources: [R3], [R8].

5.1 Purpose and Design Intent

The outer loop's job is to find the **optimal detuning** for the prevailing beam loading conditions. Rather than computing this from beam current (which would require accurate knowledge of I_b, ϕ_s, and Q_L), it uses an implicit approach: adjust the load angle offset until the cavity voltage (expressed as "strength") matches a setpoint. The physics ensures that this condition is satisfied only at the optimal operating point.

5.2 The Integrator — `subIQphaseOffsU`

C function: `subIQphaseOffsU` — `rfApp/src/db/subIQ.c,v` line 467 [R3]
EPICS record: `$(S):$(C)LOAD:ANGLE:UNADOFFS`

Regulated quantity — cavity strength:

$$\text{STRENGTH} = \frac{\text{PROBE:AMPL}}{\text{STN:VOLT}} \times 100\% \tag{5.1}$$

- `PROBE:AMPL` = cavity probe signal amplitude (kV)
- `STN:VOLT` = total station gap voltage sum across all cavities (kV)
- Clamped to 0% when `STN:VOLT < 2 kV` (hardwired floor in `subIQpowerEff`)

STRENGTH:CTRL (ao, %, DRVH=100) is the operator setpoint.
STRENGTH:DIFF (ao, %, DRVH=100) is the integrator deadband (nominal operating value **0.05%** per [R15]).

Trigger mechanism: NOT driven by the 0.5 Hz scan record. Triggered by the SNL sequencer writing PROC to LOAD:ANGLE:UNADOFFS.PROC (via phase_offset_proc PV) — see §4.3. This means the outer loop only fires when the inner loop has confirmed motor idleness and completed a full update.

Integrator update equation:

$$\theta_{unadj}[n] = \text{forget} \cdot (\theta_{unadj}[n-1] - K \cdot e_s[n-1]) \tag{5.2}$$

where:

- e_s = STRENGTH − STRENGTH:CTRL (cavity voltage strength error, %)
- K = CAVLOAD:ANGLE:K (deg/%, DRVH=1; nominal operating value **0.5 deg/%** per [R15])
- forget = CAVLOAD:ANGLE:FORGET ($\leq$1, dimensionless; nominal operating value **0.9995** per [R15])

Gate logic — two modes:

RESET (state cleared to zero) — ANY of:

1. CAVLOAD:ANGLE:CTRL ≤ 0.5 — disabled by operator (INPA)
2. Station state $\neq$ ON_CW — INPD $\neq 4$
3. Beam has MINOR or MAJOR alarm: STN:BEAM:STAT.SEVR $\{1, 2\}$ (INPC):
   - STN:BEAM:STAT is a bi record with ZSV=MAJOR — when beam is OFF, SEVR=2
   - STN:BEAM:STATINP computes: beam_curr $\geq$ RF_CUTOFF ? ON : OFF

> **Beam loss behavior [R15]:** When beam current drops below the minimum threshold, the stored load angle offset is **cleared to zero** and must regenerate from scratch as beam is re-injected. This is by design — the optimal detuning depends on beam current; a stale offset from a previous fill is not applicable to a new injection.

UPDATE (all must be true) — integrator steps: 4. Reset conditions are all false (enabled, ON_CW, beam healthy) 5. CAVLDANGERR:SUMY:SEVR.SEVR = 0 — all 4 cavities' load angle errors valid (INPB):

- This is a calc record in rf_sumy_stn.db,v with CALC="0" and four NPP MS inputs from CAV1--CAV4 LOAD:ANGLE:ERR.SEVR. Its .SEVR reflects the worst alarm across all 4 cavities via EPICS severity maximization.

6. Previous-cycle beam was healthy (INPE < 0.5) — one-cycle recovery hysteresis
7. STRENGTH.SEVR < 2.5 — strength measurement valid (INPH)
8. |strength_error| > STRENGTH:DIFF — error exceeds deadband (INPI)

Pseudo-code:

```
strength_error = STRENGTH - STRENGTH_CTRL;   // actual − desired (%)

if (NOT active):
    offset = 0.0;                            // RESET
else if (all_valid AND |strength_error| > deadband):
    offset -= strength_error * K;            // integrator step
offset *= forget;                            // leaky decay
offset = clamp(offset, -180, +180);          // hard limits
```

5.3 Total Offset — subIQphaseOffs

C function: subIQphaseOffs — rfApp/src/db/subIQ.c,v line 542 [R3]
EPICS record: $(S):$(C)LOAD:ANGLE:OFFS

```
psub->val = psub->d;                     // fixed offset (INPD)
if (psub->a > 0.5) psub->val += psub->b;  // + unadj_offset if enabled
psub->val = clamp(psub->val, psub->e, psub->c); // clip to [min, max]
```

$$\boxed{\theta_{offset} = \text{clamp}(D_{fixed} + \theta_{unadj},\ E_{min},\ C_{max})} \tag{5.3}$$

In-put	Function	In deployed SPEAR3 db
INPA	`CAVLOAD:ANGLE:CTRL` (enable)	wired
INPB	`LOAD:ANGLE:UNADOFFS` (integrator output)	wired
INPC	Max bound	absent in DB source — `#field(INPC,"10")` commented out → C = 0 on fresh start; **autosave restores 75° in normal operation** [R15]
INPD	Fixed offset	absent → D = 0
INPE	Min bound	absent → E = 0

Purpose of the fixed offset (INPD): Allows a permanent operator-set phase bias independent of the integrator. In PEP-II, this encoded a known optimal detuning from design parameters or absorbed systematic calibration offsets. In SPEAR3, INPD is absent: no fixed offset is applied.

! The outer loop defaults to disabled on a fresh IOC start without autosave.
The DB source has `#field(INPC,"10")` commented out. On a clean start without autosave, C = 0 (and E = 0), so `clamp(val, 0, 0) = 0` forces the output to zero regardless of what the integrator computes.
However, in normal SPEAR3/PEP-II operation with autosave restore, `INPC` is restored to the nominal operational value of 75° per [R15]. [O4] The official SLAC operator documentation states: *"Max Offs nominal value = 75 degrees"* and that it *"should not need to be greater than 10 degrees but is presently set higher."* With autosave active and `LOAD:ANGLE:CTRL=1`, the outer loop IS functional. The `#field(INPC,"10")` comment suppresses only the DB-compiled hard-coded default value; the 10° figure was likely a conservative initial recommendation subsequently raised to 75° in the deployed system.
Consequence of fresh start without autosave: $\theta_{offset} = 0$. The voltage integrator (`UNADOFFS`) computes values but they are discarded by the zero-valued clamp bounds.
To activate the outer loop on a fresh start, configure: `CAVLOAD:ANGLE:CTRL=1`, `STRENGTH:CTRL`, `STRENGTH:DIFF` (~0.05%), `LOAD:ANGLE:K` (~0.5 deg/%), `LOAD:ANGLE:FORGET` (~0.9995), and set `INPC` on `LOAD:ANGLE:OFFS` (max bound) to ~75°. Note also the fundamental sign-ambiguity described in §10.8 — cavity voltage is symmetric about resonance.

---

6 Sequencer: Housekeeping and Error Handling

Sources: [R1], [R2], [R11].

Source: `rfApp/src/rf_tuner_loop.st,v` (555 lines, 4 instances — `program rf_tuner_loop("STN=RRRS,CAV=X,name=CXTUNRLOOP")`)

The SNL sequencer manages the complete lifecycle of the tuner loop. §4.3 covered the normal control path within `loop_on`. This section covers startup, error handling, and state classification.

6.1 State Diagram

Figure 3 shows the five states and the transitions between them.



Figure 3: SNL tuner loop state machine. Five states; motion is commanded only from `loop_on`.

6.2 Guard Checks in loop_on

Before the normal control path (§4.3) executes, the 0.5 Hz event handler evaluates guard conditions in priority order:

Priority	Condition	Status Code	Action
1	`loop_ctrl == OFF`	`LOOP_OFF_STATUS` (1)	No motor action
2	`meas_count == 0`	`LOOP_PHASMISS_STATUS` (5)	IQA not producing data
3	Motor stuck > 5 cycles	`LOOP_SM_MOVE_STATUS` (6)	Motor alarm
4	`station_state == ON_FM`	`LOOP_ON_FM_STATUS` (8)	Wait for CW
5	Klystron power low	`LOOP_LOW_STATUS` (9)	Skip cycle
6	`dmov_meas_count < 1`	—	Wait for motor idle
7	Motor missed target	`LOOP_SM_CTRL_STATUS` (7)	Sanity alarm

Only if all guards pass does the normal path fire (§4.3 Step 3).

6.3 Measurement Accumulator

A separate `when(efTest(meas_ready_ef))` block counts measurement events between loop ticks:

```
meas_count++
if (sm_dmov == SM_DONE_MOVING):    dmov_meas_count++
else:                               dmov_meas_count = 0
```

The main control path requires `dmov_meas_count` $\geq$ `LOOP_DMOV_MEAS (=1)` — at least one measurement where the motor was idle — before acting.

6.4 State Variables and Persistent State

Understanding which variables carry memory across cycles is critical for the upgrade design:

Control state (persistent memory — the loop's "remembrance"):

Variable	PV	Memory type
`posn_ctrl`	`TUNR:POSN:CTRL`	**Inner loop state.** The absolute motor position command. Updated each cycle as $x_{actual} + \Delta x$.
`LOAD:ANGLE: UNADOFFS .VAL`	`CAV{N}LOAD:ANGLE: UNADOFFS`	**Outer loop state.** Accumulated strength-error-based corrections with leaky forgetting. Stored in EPICS record `.VAL`.

Sequencer bookkeeping (SNL-internal):

Variable	Role
`loop_state`	Current SNL state (init/off/reset/on)
`loop_status`	Last computed status code
`prev_loop_ctrl`	Previous cycle's enable value (edge detection + sanity check)
`prev_loop_ status`	Last status written to PV (change-detect)
`meas_count`	Measurement events since the last loop tick
`dmov_meas_count`	Motor-idle measurements (gating condition)
`nomov_count`	Consecutive stuck-motor cycles

Measured values (stateless — recomputed each cycle):

Variable	Source
`posn_delta`	`subIQphase2posn` output
`sm_posn`	Motor record `.RBV`
`sm_dmov`	Motor record `.DMOV`
`station_state`	`STN:STATE:RBCK`
`load_angle_sevr`	`LOAD:ANGLE:ERR.SEVR`
`klys_frwd_pwr`	`KLYSOUTFRWD:POWER`

---

7 Disabled Path: Park Mode (PEP-II Only)

Sources: [R4], [R11], [R14].

These functions existed for PEP-II (a collider with two rings). In SPEAR3 they are present in the code but permanently disabled. For PEP-II, the nominal park frequency offset was $\pm$**340 kHz** per station: half the cavities were parked at +340 kHz relative to resonance and the other half at -340 kHz [R15].

7.1 Frequency Offset Estimation

C function: `subSysFreqOff` — `rfApp/src/db/subSys.c,v` line 115 [R4]
EPICS record: `$(S):$(C):FREQ:OFFS`

Estimates frequency offset from tuner position using a polynomial:

$$f_{offset} = p_0 + p_1 \cdot \Delta x + p_2 \cdot \Delta x^2 + p_3 \cdot \Delta x^3 + t_1 \cdot V_{cav}^2 \tag{7.1}$$

where Δx = Tuner Posn − ON Home Posn (mm) — position relative to the ON mode home (`TUNR:POSN:ONHOME`), **not** absolute position. Parameters are calibrated per cavity and thermal conditions. The $t_1 \cdot V_{cav}^2$ term accounts for RF-heating-induced detuning. The polynomial is typically accurate to **~10 kHz** per [R15].

Nominal coefficient ranges (from [R15]):

Coefficient	Nominal Range	Units
p_0	10 to 20	kHz
p_1	20 to 30	kHz/mm
p_2	0.4 to 0.6	kHz/mm^2
p_3	−0.01 to +0.01	kHz/mm^3
t_1	≈ -0.00010	kHz/kV^2

Output smoothing: A first-order IIR smoother with $S = 0.5$ is applied to the frequency offset output before it reaches `FREQ:ERR` [R15]:

$$f_{smooth}[n] = (1 - S) \cdot f_{offset}[n] + S \cdot f_{smooth}[n-1] \tag{7.2}$$

This halves the effective bandwidth of the polynomial estimate, attenuating noise that could cause excessive motor movement in PARK mode.

7.2 Park Frequency Error

C function: `subSysFreqErr` — `rfApp/src/db/subSys.c,v` line 147 [R4]
EPICS record: `$(S):$(C):FREQ:ERR`

$$\text{FREQ:ERR} = 0.00004726891 \cdot Q_L \cdot (f_{desired} - f_{offset}) \quad [\text{degrees}] \tag{7.3}$$

This converts a frequency error to phase error in degrees so `subIQphase2posn` can handle both ON_CW (phase) and PARK (frequency) paths.

Disabled in SPEAR3: `INPB` ($Q_L \approx 7000$) is commented out → B=0 → output always 0.

```
grecord(sub,"$(S):$(C):FREQ:ERR") {
    field(SNAM,"subSysFreqErr")
    field(INPA,"0.00004726891")   # 90/(476e3 × 4) deg/kHz
  # field(INPB,"7000")            # Q_L --- COMMENTED OUT
    field(INPD,"$(S):$(C):FREQ:OFFS.VAL  NPP MS")
}
```

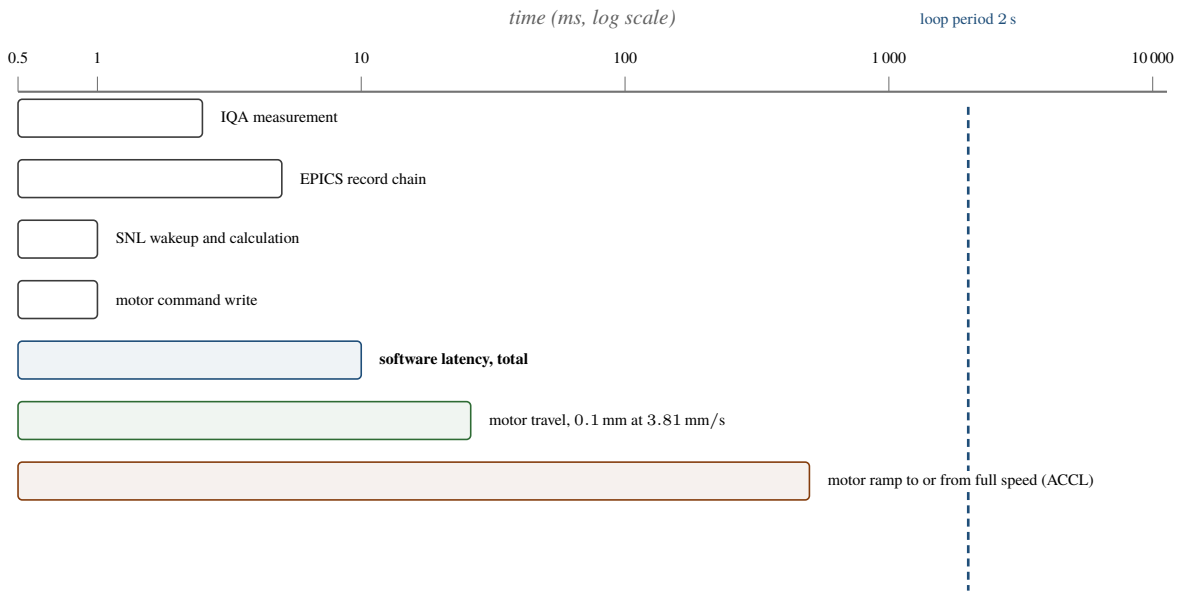


Figure 4: Timing of one tuner loop cycle. Every contribution, mechanical ones included, sits inside the 2 s loop period; the software terms are three decades below it.

Part III
Analysis

8 Control Loop Analysis

8.1 Loop Timing Budget

Figure 4 places each contribution to one loop cycle on a common logarithmic time axis. The ratios are what matter: everything in software is negligible against the 2 s period, and the mechanical ramp is the only term of comparable size.

8.2 Open-Loop Gain (Proportional Path)

The linearized loop gain from phase error to stepper position change per cycle:

$$G_{OL} = C_{mm/deg} \cdot G_{loop} = 0.020\text{–}0.030 \ \frac{\text{mm}}{^\circ} \cdot G_{loop} \tag{8.1}$$

A position change of Δx mm changes cavity resonant frequency by:

$$\frac{df}{dx} \approx p_1 \quad [\text{kHz/mm}] \tag{8.2}$$

(the linear coefficient of the tuner polynomial, calibrated per cavity)
Effective loop gain (phase domain, per cycle with $G_{loop} = 1$):

$$G_{eff} = C_{mm/deg} \cdot p_1 \cdot \frac{90^\circ}{4 f_0} \cdot Q_L \tag{8.3}$$

For nominal values ($C = 0.020$ mm/° (lower end of range), $p_1 \approx 10$ kHz/mm, $Q_L \approx 7000$, $f_0 = 476$ MHz):

$$G_{eff} \approx 0.02 \times 10 \times 0.00004726891 \times 10^3 \times 7000 \approx 66 \tag{8.4}$$

Note: Equation (8.4) is the single-cycle proportional gain. The 0.5 Hz rate and motor velocity limit prevent instability by severely band-limiting the loop. `CAVTUNR:LOOP:GAIN` (0–1) is critical for stability.
The factor $90°/4f_0$ in Eq. (8.3) is the constant `0.00004726891` that `subSysFreqErr` uses [R4]. That routine documents it as the *park* frequency-error scaling (degrees per kHz per unit Q_L), which is about five times smaller than the physical load-angle conversion in Eq. (2.4). Whether the inner-loop gain estimate should use the park constant or the physical one has not been resolved here; the qualitative conclusion — $G_{eff} \gg 1$, so the loop gain scalar is what keeps the loop stable — holds either way. **[O7]**

8.3 Integrator Dynamics (Outer Loop)

The load angle offset integrator is a **leaky first-order IIR integrator**, Eq. (5.2). Its two tuning constants set the DC gain and the leak rate.

K gain (`CAVLOAD:ANGLE:K`, deg/%): Per-sample step size. `DRVH=1` limits K ≤ 1 deg/%.

Forgetting factor (`CAVLOAD:ANGLE:FORGET`, 0 < forget ≤ 1): Makes the integrator leaky:

- **DC gain:** $G_{DC} = -K/(1 - \text{forget})$ deg/%
- **Decay time constant:** $\tau = -T/\ln(\text{forget})$ where $T = 2$ s is the loop period
- If the strength error disappears, the accumulated offset decays to zero with time constant τ

`forget`	τ (seconds)	DC gain (K=1 deg/%)	Notes
1.000	∞ (pure integrator)	∞	Winds up indefinitely
0.999	≈ 2000 s (33 min)	−1000 deg/%	Very slow, high-gain
0.990	≈ 200 s (3.3 min)	−100 deg/%	Moderate — typical range
0.900	≈ 19 s	−10 deg/%	Aggressive

Configuration note: Both K and FORGET are `ao` records with `PINI=YES` but no hardcoded defaults. On a fresh start without autosave, both = 0, silently disabling the integrator.

8.4 Bandwidth Hierarchy and Disturbance Rejection

- Inner proportional loop bandwidth: ~0.01–0.05 Hz (operator-tunable via `LOOP:GAIN`)
- Outer integrator loop bandwidth: ~0.0001–0.008 Hz (set by K and forget; $f \approx 1/2\pi\tau$)

Primary disturbance — beam loading: The dominant source of cavity detuning. As beam is injected, decays, or is lost, $\Delta f \propto I_{b1} \sin\phi_s / V_{gap}$ (Eq. (2.1)) shifts the cavity resonance. Under normal SPEAR3 operation (slow current decay, hours lifetime), the average detuning rate is slow enough for the 0.5 Hz loop. However, discrete events — injection bursts, fast beam loss — can shift detuning faster than the 2 s loop period.

Secondary disturbance — thermal drift: RF heating shifts resonant frequency on minutes-to-hours timescale. Much smaller than beam loading at SPEAR3 gradients. Easily handled by the inner loop.

Uncorrected disturbances: Fast detuning transients (injection events, beam loss) and components > ~0.25 Hz are not corrected by the 0.5 Hz loop. These manifest as transient phase errors that ring down over several loop cycles.

9 Hardware Interface

Sources: [R6], [R7], [R8], [R12], [R16].

9.1 Stepper Motor Record

EPICS record type: `steppermotor` (custom AB 1746-HSTP1 driver record)
Physical device (legacy): Allen-Bradley 1746-HSTP1 stepper module
Physical device (built and tested, not installed): Galil DMC-4143 motion controller, to be driven by LLRF9 Unit 1 **[O6]**

```
grecord(steppermotor,"$(S):$(C)TUNR:STEP:MOTOR") {
    field(DTYP,"AB-1746HSTP1")
    field(OUT,"#L0 A2 C$(M) S0 @@0x8413 0x0010 500")
    field(DOL,"$(S):$(C)TUNR:POSN:CTRL.VAL  NPP NMS")
    field(OMSL,"closed_loop")
    field(ACCL,".5")          # seconds to reach velocity
    field(VELO,"3")           # velocity: 3 rev/s = 3.81 mm/s
    field(DIST,"0.003175")    # mm per step
    field(MODE,"Position")
    field(CMOD,"Position")
    field(MRES,"400")         # steps per revolution
    field(PREC,"3")
    field(EGU,"mm")
    field(DRVH,"18")          # upper drive limit: +18 mm
    field(DRVL,"-29.5")       # lower drive limit: -29.5 mm
    field(RDBD,"0.015875")    # retry deadband: 5 steps
}
```

9.2 Mechanism Specifications

Parameter	Value	Calculation
Step resolution	0.003175 mm/step	1.27 mm/rev ÷ 400 steps/rev
Retry deadband	0.015875 mm	5 steps exactly
Maximum velocity	3.81 mm/s	VELO 3 rev/s × MRES 400 steps/rev × DIST 0.003175 mm/step
Acceleration	7.62 mm/s^2	ACCL = 0.5 s to reach full velocity
Travel range	47.5 mm total	-29.5 to +18 mm
Time to traverse full range	~12.5 s	47.5 mm at 3.81 mm/s
Time to stop from full speed	0.5 s	ACCL is specified directly as seconds to velocity
Min detectable move	0.015875 mm	1 RDBD = 5 steps

These are the *commanded* limits taken from the motor record [R8]. The June 2023 tuner inspection recorded bellows and mechanical-stop problems on cavities A and B and a limit-switch problem on cavity C. In September 2026 J. Sebek and R. Lion found that the plungers move only about ± 4 mm over a year, against limit switches set at about ± 25 mm; the mechanical travel will therefore be restricted to ± 10 mm, which does not stress the bellows [R17].

9.3 Position Control and Initialization

The motor uses **absolute position mode**. On IOC startup:
1. `TUNR:STEP:CHCKINIT` verifies motor is initialized
2. `TUNR:STEP:INIT` sequence captures current hardware position → `TUNR:POSN:CTRL`
3. Subsequent commands write to `TUNR:POSN:CTRL`, which FLNKs directly to the motor record

---

10 Design Gaps and Upgrade Opportunities

Sources: [R8], [R15], [R17].

10.1 Commented-Out Parameters Make Tuner Non-Functional on Fresh Start

The three key parameters for `subIQphase2posn` (deadband B, conversion C, max-delta D) are commented out in the database. On a clean start without autosave restore, C=0 → no motor movement, D=0 → output clamped to 0.

Recommendation: Encode these as `ao` records with `PINI=YES` and default values.

10.2 Outer Loop Disabled on Fresh Start (Functional with Autosave)

The LOAD:ANGLE:OFFS bounds (INPC) are commented out in the DB source (`#field(INPC,"10")`): `clamp(val, 0, 0) = 0` on a fresh IOC start without autosave. In normal operation with autosave restore, INPC is restored to 75° [R15], making the outer loop potentially active when `LOAD:ANGLE:CTRL=1`. Additionally, K and FORGET have no hardcoded DB defaults (both = 0 on fresh start without autosave).

Risk: On a fresh IOC start without autosave (disaster recovery, new deployment), the outer loop is silently non-functional. An operator setting `LOAD:ANGLE:CTRL=1` may believe the outer loop is active when it is not.

Recommendation: Encode explicit defaults for K (0.5 deg/%), FORGET (0.9995), and OFFS bounds (±75°) as DB `PINI=YES` values in the upgrade design.

10.3 No Feedforward for Beam Loading

Beam loading is the primary disturbance (§2.1, §8.4). The only compensation path is the 0.5 Hz reactive inner loop. The outer integrator was designed to find optimal detuning implicitly, but is inactive. Even if active, its ~50 s time constant cannot respond to injection transients.

There is no feedforward from beam current to detuning setpoint. This leaves the cavity transiently detuned during injection and loss events.

Upgrade opportunity: Real-time detuning feedforward from DCCT beam current, using Eq. (2.1) directly:

$$\Delta f_{ff} = -\frac{f_0}{2} \cdot \frac{R}{Q} \cdot \frac{I_{b1} \sin \phi_s}{V_{gap}} \tag{10.1}$$

Computable in the LLRF9 with sub-millisecond latency.

10.4 Implicit vs. Explicit Physics

The legacy system uses an implicit integrator to find optimal detuning rather than first-principles computation ($\psi_{opt} = \arctan(2Q_L\Delta\omega/\omega_0)$). This is robust but slow.

Upgrade opportunity: Explicit model-based Δf_{opt} from measured I_b, V_{gap}, and ϕ_s — directly enabled by LLRF9 DSP.

10.5 Motor Resolution vs. Phase Resolution

With C = 0.020 mm/° (minimum end of nominal range; worst-case resolution):

- 1 step (0.003175 mm) 0.159° phase change
- RDBD (5 steps = 0.015875 mm) 0.794° phase

For $Q_L \approx 7000$: 0.159° corresponds to $\Delta f \approx 95$ Hz minimum resolvable correction. The Galil controller would achieve sub-step positioning once commissioned.

10.6 No State Estimation or Predictive Control

Purely reactive — responds to measured error with no prediction. The 0.5 Hz rate cannot correct disturbances above its 0.25 Hz Nyquist limit (AC ripple, beam chopping).

10.7 Outer Loop Trigger Is SNL-Mediated

The outer integrator is triggered by the SNL writing PROC, not by the scan record. If the sequencer is not running (IOC restart, task crash) or the motor is persistently stuck (`DMOV=0`), the outer loop freezes silently.

Recommendation: Consider independent scan-based outer loop in upgrade.

10.8 Voltage Loop Sign Ambiguity — Lorentzian Symmetry Problem

This is a fundamental control-design vulnerability in the outer (voltage) loop.

The cavity voltage response vs. detuning is a Lorentzian:

$$V(\Delta f) = \frac{V_{max}}{\sqrt{1 + \left(\frac{2Q_L\,\Delta f}{f_0}\right)^2}} \tag{10.2}$$

This curve is **symmetric around resonance** — the same reduced voltage appears at $+\Delta f$ and $-\Delta f$. The integrator uses voltage error alone (`STRENGTH − STRENGTH:CTRL`) to decide which direction to push the phase offset:

```
psub->val -= psub->l * psub->j;   // offset -= (actual − desired) × K
```

When voltage is too low (actual < desired), the integrator always **increases** the offset. This is correct only if the cavity is detuned on one side of resonance. If it happens to be on the other side:

```
Scenario: cavity detuned ABOVE resonance (Δf > 0)
  → voltage < target  (down the Lorentzian slope)
  → error = actual − desired < 0
  → offset -= (negative × K) → offset INCREASES
  → inner loop drives tuner further above resonance
  → voltage drops MORE → error grows → offset increases faster
  → POSITIVE FEEDBACK → runaway to ±180° clamp
```

The integrator converges only from the **correct side** of resonance; from the wrong side it diverges. Which side is "correct" depends on the sign of K and the tuner geometry (sign of df/dx), but the fundamental issue is that voltage alone does not distinguish $+\Delta f$ from $-\Delta f$.

Why this doesn't cause problems in practice: In SPEAR3 as deployed, the outer loop is non-functional on fresh start (see §5.3, §10.2). The inner loop uses phase error, which is **monotonic** with detuning (not symmetric), so it does not have this ambiguity. If the outer loop were ever activated (e.g., via autosave with Max Offs = 75°):

1. **Normal operation (beam loading dominant):** Beam loading consistently pulls the cavity to one side of resonance (determined by $\sin\phi_s$). If the initial sign convention is chosen for this side, convergence is reliable under steady-state conditions.
2. **Transient risk:** After a beam loss/injection event, the cavity could briefly end up on the wrong side of resonance. The integrator would push in the wrong direction until either (a) the inner loop's phase control pulls it back, or (b) the offset hits the $\pm 180°$ clamp.
3. **Startup ambiguity:** On first activation, the cavity's initial detuning direction is unknown. The integrator has a 50/50 chance of guessing wrong.

Upgrade recommendation: Any replacement for the voltage loop should use **signed detuning information** (available from the phase measurement) rather than unsigned voltage error. Options include:

- Feedforward from beam current (§10.3) — eliminates the need for voltage feedback entirely
- Model-based optimal detuning (§10.4) — computes Δf_{opt} directly from I_b, V_{gap}, ϕ_s
- If a voltage feedback loop is retained, include sign discrimination by checking the inner loop's phase error sign to determine which side of resonance the cavity is on, and flip the integrator sign accordingly

---

11 Operational Summary: SPEAR3 As Deployed

In SPEAR3 as deployed, the tuner operates exclusively in pure resonance-seeking mode:

1. **Inner loop (active):** 0.5 Hz proportional phase controller. Error = `PROBE:PHASE.L` − `FRWD:PHASE.L`. Motor drives to nullify this error → cavity stays at the calibrated resonant frequency.
2. **Outer loop (inactive on fresh start; potentially active with autosave):** The `LOAD:ANGLE:OFFS` max bound (INPC) is commented out in the DB source — on a fresh start `clamp(val, 0, 0) = 0` discards the integrator output. In normal SPEAR3 operation, autosave restores INPC to 75°, allowing the outer loop to function when `LOAD:ANGLE:CTRL=1`. In practice at SPEAR3 with modest beam loading the outer loop has always run at $\theta_{offset} = 0$ (pure resonance-seeking mode). See §5.3 and §10.2.
3. **PARK mode (disabled):** Permanently disabled via commented-out `INPB` in `FREQ:ERR`.

What "resonance" means here: The calibrated zero reference from the no-beam IQA calibration procedure (§2.4). The system keeps `probe_phase − fwd_phase = 0`, which equals resonance only if the calibration is valid.

What is NOT compensated: With $\theta_{offset} = 0$, the tuner always drives toward zero detuning (resonance). It does not apply the optimal detuning angle that would minimize generator power under beam loading. For SPEAR3's modest beam loading this is acceptable. For higher beam loading conditions, the outer voltage loop exists in the architecture but requires explicit configuration to activate.

A Complete EPICS Signal Chain

A.1 Data Flow (SPEAR3 ON_CW Mode)

```
Step 1: IQA hardware acquires I,Q at 400 Hz
  SRF1:CAV1PROBE:IQ → subIQphase → SRF1:CAV1PROBE:PHASE  (degrees, unadjusted in .L)
  SRF1:CAV1FRWD:IQ  → subIQphase → SRF1:CAV1FRWD:PHASE   (degrees, unadjusted in .L)

Step 2: Load angle target (slow integrator, if enabled)
  SRF1:CAV1:STRENGTH (= probe_amplitude / station_voltage × 100%)
  SRF1:CAV1:STRENGTH:CTRL (operator setpoint, %)
  → SRF1:CAV1LOAD:ANGLE:UNADOFFS (subIQphaseOffsU, triggered by SNL)
  → SRF1:CAV1LOAD:ANGLE:OFFS     (subIQphaseOffs, = fixed_offs + unadj_offs, bounded)

Step 3: Load angle error
  SRF1:CAV1LOAD:ANGLE:ERR (subIQphaseErr):
    error = PROBE:PHASE.L − FRWD:PHASE.L + LOAD:ANGLE:OFFS
    wrapped to [−180°, +180°]

Step 4: Phase error to position delta
  SRF1:CAV1TUNR:POSN:DELTA (subIQphase2posn):
    if |error| > deadband_B:
        delta = error × C_mm_per_deg × gain_E
        delta = clamp(delta, −D_mm, +D_mm)
    else:
        delta = 0
  → fires event SRF1:CAVTUNR:LOOP:READY

Step 5: SNL sequencer (one instance per cavity)
  Wakes on READY event
  new_posn = sm_posn + posn_delta
  new_posn = clamp(new_posn, sm_drvl, sm_drvh)
  writes new_posn → SRF1:CAV1TUNR:POSN:CTRL

Step 6: Motor response
  SRF1:CAV1TUNR:POSN:CTRL FLNK → SRF1:CAV1TUNR:STEP:MOTOR
  Motor moves to absolute position new_posn
  Readback: SRF1:CAV1TUNR:POSN (motor .RBV → EPICS ao)
```

A.2 Record Trigger Chain

```
SRF1:CAVTUNR:LOOP:SEQ  [SCAN="2 second"]
    → fires FLNK → SRF1:CAVTUNR:LOOP:READY (event)
    → SRF1:CAV{1,2,3,4}TUNR:POSN:DELTA processes (event-triggered)
        → each POSN:DELTA FLNK → SRF1:CAV{N}TUNR:LOOPMEAS:READY event
            → wakes SNL sequencer instance (per cavity)
                → SNL reads posn_delta, updates motor: pvPut(TUNR:POSN:CTRL)
                → SNL writes PROC to LOAD:ANGLE:UNADOFFS (via phase_offset_proc)
                    → subIQphaseOffsU fires → updates voltage integrator
                    → FLNK → LOAD:ANGLE:OFFS recomputed
                        → feeds next cycle's LOAD:ANGLE:ERR
```

B Parameter Reference

B.1 Operator-Settable Parameters (Station-Level)

These are ao/bo records in `rf_stn_cav.db,v`, shared across all cavities (prefix `$(S):CAV`):

PV (with prefix `SRF1:CAV`)	Type	EGU	DRVH	Default	Description
`TUNR:LOOP:CTRL`	bo	—	—	—	Master loop enable (0=off, 1=on)
`TUNR:LOOP:GAIN`	ao	—	1	autosave	Phase-to-position gain scalar (0–1)
`LOAD:ANGLE:K`	ao	deg/%	1	autosave	Integrator K constant
`LOAD:ANGLE:FORGET`	ao	—	1	autosave	Forgetting factor ($\leq$1)
`LOAD:ANGLE:ERRMAX`	ao	deg	180	autosave	Max allowed load angle error
`LOAD:ANGLE:CTRL`	bo	—	—	autosave	Enable load angle offset integrator
`STRENGTH:CTRL`	ao	%	100	autosave	Desired cavity strength setpoint (%)
`STRENGTH:DIFF`	ao	%	100	autosave	Voltage integrator deadband

B.2 Per-Cavity Parameters (rf_cav.db,v)

PV (e.g., `SRF1:CAV1`)	Function	Default	Notes
`TUNR:POSN:ONHOME`	Home position for freq polynomial	autosave	Reference for `subSysFreqOff`
`TUNR:POSN:PARKHOME`	Park home position	autosave	PEP-II only
`LOAD:ANGLE:OFFS`	Total load angle offset (computed)	0	output of `subIQphaseOffs`

B.3 Commented-Out Parameters (Require Autosave)

Sub Record	Input	Function	Design Value	Status
`TUNR:POSN:DELTA`	INPB	Phase deadband (°)	0.25	`#field(INPB,"0.25")` — commented
`TUNR:POSN:DELTA`	INPC	mm per degree	0.020–0.030 range	`#field(INPC,"0.02")` — commented
`TUNR:POSN:DELTA`	INPD	Max delta per cycle (mm)	1.0	`#field(INPD,"1")` — commented
`LOAD:ANGLE:OFFS`	INPC	Max offset bound (°)	10 (DB); **75 operational** [R15]	`#field(INPC,"10")` — commented; autosave restores 75° in normal operation
`FREQ:ERR`	INPB	Loaded cavity Q	7000	`#field(INPB,"7000")` — commented

B.4 Physical Constants (Hardwired)

Constant	Value	Location	Meaning
`90/(476e3 × 4)`	0.00004726891 deg/kHz	`rf_cav.db,v`, FREQ:ERR INPA	Phase per unit frequency
RF frequency	476 MHz	Implicit	SPEAR3 accelerating frequency
Stepper DIST	0.003175 mm/step	Motor record	Linear resolution
Stepper MRES	400 steps/rev	Motor record	Angular resolution
Stepper RDBD	0.015875 mm	Motor record	5-step retry deadband
Motor DRVH/DRVL	+18 / −29.5 mm	Motor record	Travel limits
Motor VELO	3 rev/s (= 3.81 mm/s)	Motor record	Maximum velocity
Motor ACCL	0.5 s	Motor record	Seconds to reach velocity (= 7.62 mm/s^2)

C SNL Variables, Constants, and Status Codes

C.1 Complete PV Binding Table

SNL Variable	Type	EPICS PV	Role
`loop_ctrl`	int	`{STN}:CAVTUNR:LOOP:CTRL`	Master enable, monitored (bo)
`loop_state`	int	`{STN}:CAV{CAV}TUNR:LOOP:STATE`	Output: current SNL state code
`loop_status`	int	`{STN}:CAV{CAV}TUNR:LOOP:STATUS`	Output: status code
`loop_status_string_c`	string	`{STN}:CAV{CAV}TUNR:LOOP:STRING`	Output: human-readable status text
`station_state`	int	`{STN}:STN:STATE:RBCK`	Monitored, drives state branching
`posn_ctrl`	float	`{STN}:CAV{CAV}TUNR:POSN:CTRL`	Motor position command (mm)
`posn_delta`	float	`{STN}:CAV{CAV}TUNR:POSN:DELTA`	Per-cycle position increment
`posn_new`	float	`{STN}:CAV{CAV}TUNR:POSN:LOOP`	Logged: newly computed target
`posn`	float	`{STN}:CAV{CAV}TUNR:POSN`	Potentiometer position readback
`posn_on_home`	float	`{STN}:CAV{CAV}TUNR:POSN:ONHOME`	Home for ON mode
`posn_park_home`	float	`{STN}:CAV{CAV}TUNR:POSN:PARKHOME`	Home for PARK (PEP-II)
`posn_mdel`	float	`{STN}:CAV{CAV}TUNR:POSN.MDEL`	Monitor delta
`sm_posn`	float	`{STN}:CAV{CAV}TUNR:STEP:MOTOR.RBV`	Motor readback (authoritative)
`sm_dmov`	int	`{STN}:CAV{CAV}TUNR:STEP:MOTOR.DMOV`	Done-moving flag, monitored
`sm_drvh`	float	`{STN}:CAV{CAV}TUNR:STEP:MOTOR.DRVH`	Upper travel limit

SNL Variable	Type	EPICS PV	Role
sm_drvl	float	{STN}:CAV{CAV}TUNR:STEP:MOTOR.DRVL	Lower travel limit
sm_rdbd	float	{STN}:CAV{CAV}TUNR:STEP:MOTOR.RDBD	Retry deadband
phase_offset_proc	int	{STN}:CAV{CAV}LOAD:ANGLE:UNADOFFS.PROC	Outer loop trigger
load_angle_sevr	int	{STN}:CAV{CAV}LOAD:ANGLE:ERR.SEVR	Load angle alarm severity
klys_frwd_pwr	float	{STN}:KLYSOUTFRWD:POWER	Klystron forward power
klys_frwd_pwr_min	float	{STN}:KLYSOUTFRWD:POWER:MIN	Minimum power threshold
loop_ready	int	{STN}:CAVTUNR:LOOP:READY	Station-wide loop event (0.5 Hz)
meas_ready	int	{STN}:CAV{CAV}TUNR:LOOPMEAS:READY	Per-cavity measurement event

C.2 Internal Counters (No PV Binding)

Variable	Purpose
meas_count	LOOPMEAS:READY events since last LOOP:READY
dmov_meas_count	Motor-idle measurements (gating condition)
nomov_count	Consecutive stuck-motor cycles
nonfunc_count	Consecutive bad cycles (rate-limits alarms)
reset_count	Reset attempt counter
prev_loop_ctrl	Previous enable value (edge detection + sanity check)
prev_loop_status	Previous status code (change-detect)

C.3 Constants (from `rf_tuner_loop_defs.h,v`)

Constant	Value	Description
LOOP_DMOV_MEAS	1	Min done-moving measurements before acting
LOOP_NOMOV_COUNT	5	Stuck-motor cycles before alarm
LOOP_NONFUNC_INTERVAL	1	Bad cycles before NONFUNC alarm
LOOP_RESET_COUNT	5	Max reset attempts
LOOP_MAX_DELAY	60	Max startup wait (seconds)
SM_DONE_MOVING	1	DMOV value = done

C.4 Status Codes

Code	Name	Meaning
0	LOOP_UNKNOWN_STATUS	Initial / unresolved
1	LOOP_OFF_STATUS	Disabled by operator
2	LOOP_GOOD_STATUS	Normal operation
3	LOOP_RESET_STATUS	Performing reset
4	LOOP_HOME_STATUS	Moving to home
5	LOOP_PHASMISS_STATUS	No IQA data
6	LOOP_SM_MOVE_STATUS	Motor stuck
7	LOOP_SM_CTRL_STATUS	Motor missed target
8	LOOP_ON_FM_STATUS	Station in FM
9	LOOP_LOW_STATUS	Klystron power low
13	LOOP_LDANGLIM_STATUS	Load angle at limit

D Source Reference Index

Every file below was located and read in the repository working copy; the line numbers are from the head revision of each RCS ,v archive.

D.1 Primary Source Code

Ref	File	Location	Lines	Role
[R1]	rf_tuner_loop.st,v	rfApp/src/seq/	555	Tuner loop SNL state machine — five states, four instances
[R2]	rf_tuner_loop_defs.h,v	rfApp/src/seq/	—	Loop constants, mode codes and the 14 status codes
[R3]	subIQ.c,v	rfApp/src/db/	966	subIQphaseOffsU (L467), subIQphaseOffs (L542), subIQphaseErr (L566), subIQphase2posn (L586)
[R4]	subSys.c,v	rfApp/src/db/	465	subSysFreqOff (L115), subSysFreqErr (L147) — PARK mode only
[R5]	rf_station_state.h,v	rfApp/src/db/	—	Station state codes; ON_CW = 4
[R6]	steppermotorRecord.dbd,v	stepper/stepper/	—	Field definitions: VELO *Velocity Rotation/Sec*, ACCL *Seconds to Velocity*, MRES *Motor Pulses/Revolution*, DIST *Dist of One Pulse*
[R7]	steppermotorRecord.c,v	stepper/stepper/	—	Record support: velocity = velo*mres, acceleration = (1/accl)*velo*mres

D.2 EPICS Database Files

Ref	File	Location	Content used here
[R8]	`rf_cav.db,v`	`rfApp/Db/`	Per-cavity records. `subIQphase2posn` record at L560 with INPB/INPC/INPD commented out; `steppermotor` record at L578 (ACCL .5, VELO 3, DIST 0.003175, MRES 400, DRVH 18, DRVL -29.5, RDBD 0.015875)
[R9]	`rf_stn_cav.db,v`	`rfApp/Db/`	Station-level tuner records; `$(S):CAVTUNR:LOOP:SEQ` carries `SCAN="2 second"` — the loop cadence
[R10]	`rf_stn_4CV.substitutions,v`	`rfApp/Db/`	Four-cavity station instantiation; station prefix `SRF1:`

D.3 Analysis Notes and Companion Documents

Ref	Source	Content used here
[R11]	`codeReviewTechnicalNotes/05-snl-state-machines.md`	SNL state machine inventory; PARK mode disabled
[R12]	`codeReviewTechnicalNotes/06-plc-stepper-motors.md`	Stepper motor topology; Galil planned as the AB replacement
[R13]	`codeReviewTechnicalNotes/07-epics-databases.md`	Station-level parameter inventory
[R14]	`codeReviewTechnicalNotes/08-signal-processing.md`	Polynomial frequency offset and smoothing
[R15]	SSRL RF operator display, **Tuner Inputs & Constants** panel	Nominal *operational* values for the parameters that are commented out in the database source (deadband, mm/degree conversion, loop gain, integrator K and forget factor, `LOAD:ANGLE:OFFS` bounds, polynomial coefficient ranges). The screen file itself is not archived in this repository **[O1]**
[R16]	`llrf/tuners/galil/functioningGalil20250825SwapABToManual.txt`	August 2025 Galil test moves on the existing cavity tuners
[R17]	Doc L — SPEAR3 RF System Legacy Architecture	System-level context; stepper velocity and tuner mechanism findings
[R18]	Doc P — RF Physics and Plant	Beam-loading detuning, optimum load angle, R/Q